

TASK 1

VOICE ASSISTANCE

OBJECTIVE; To Build a Python-based voice assistant that listens to spoken commands and responds with useful actions. Beginners implement core voice recognition; advance builds add NLP (NATURAL LANGUAGE PROCESSING), API integrations, and smart home features.

Here, is the steps to implement the Python based voice.

ACTUALLY,

- ✓ What the project should do.?
- ✓ The Python Assistant will:
- ✓ Take voice input through microphone.
 - Respond to HELLO
 - Tell the current time and date
 - Search something on the WEB
 - Ask to repeat if it doesn't understand
 - Speak all response using text-to -speech

```
import speech recognition as sr
```

```
import pyttsx3
```

```
import datetime
```

```
import web browser
```

```
# Text-to-speech engine
```

```
engine = pyttsx3.init()
```

```
# Function to speak
```

```
def speak(text):
```

```
    print ("Assistant:", text)
```

```
    engine. say(text)
```

```
engine. Run And Wait ()
```

```
# Function to listen
```

```
def listen ():
```

```
    recognizer = sr. Recognizer ()
```

```
    with sr. Microphone () as source:
```

```
        print ("Listening...")
```

```
        recognizer.adjust_for_ambient_noise(source)
```

```
    try:
```

```
        audio = recognizer. Listen (source)
```

```
        command = recognizer. Recognize _goo
```

```
        print ("You:", command)
```

```
        return command. Lower ()
```

```
    except sr. Unknown Value Error:
```

```
        speak ("Sorry, I could not understand. Please repeat.")
```

```
        return ""
```

```
    except sr. Request Error:
```

```
        speak ("Sorry, there is a problem with the speech service.")
```

```
        return ""
```

```
# Main voice assistant
```

```
def voice _assistant ():
```

```
    speak ("Hello! I am your voice assistant. How can I help you?")
```

```
while True:
command = listen ()
if command == "":
    continue
# Greeting
if "hello" in command or "hi" in command:
    speak ("Hello! Nice to meet you.")

# Time
elif "time" in command:
    current _time = datetime. datetime. Now (). Strf time ("%I: %M %p")
    speak ("The current time is " + current _time)

# Date
elif "date" in command:
    current _date = datetime. datetime. Now (). Strf time ("%d %B %Y")
    speak ("Today's date is " + current _date)

# Web search
Elif  "search" in command:
    search _query = command. Replace ("search", ""). Strip ()
if search _query:
    speak ("Searching for " + search _query)
```

```
web browser. Open (  
    "https://www.google.com/search?q=" + search_query  
)  
else:  
    speak ("Please tell me what you want to search.")  
  
# Exit  
elif "exit" in command or "stop" in command or "bye" in command:  
    speak ("Goodbye! Have a nice day.")  
    break  
  
else:  
    speak ("Sorry, I don't understand that command.")  
  
# Run the assistant  
Voice_assistant ()  
  
Exit;  
  
Return;
```